

Tribhuvan University
Faculty of Humanities and Social Sciences
Patan Multiple Campus Mid-Term

2026

Masters in Computer Application (MCA)

Course Title: Advanced Database Management System

Code No.: MCA 503

Full Marks: 60

Pass Marks: 30

Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt Any Six Questions:

[6 x 5 = 30]

1. What is object and literal in object-oriented database? Compare object identifier (OID) with the primary key in the database.
2. How an enhanced entity relationship (EER) Schema is mapped into an object database (ODB) Schema? Explain.
3. What is query processing? Explain the various steps involved in high level query processing by the database management system.
4. Define parallel query processing and explain how parallelism can be achieved at the levels of individual operators, within a single query, and among multiple queries.
5. How incremental view maintenance contributes to the optimization? Explain the incremental view maintenance for join operation?
6. What is cost based query optimization? List the various catalog information stored in database catalog and used by the query optimizer to estimate the costs of various execution strategies.
7. What is transparency in the context of distributed database systems? Explain the various category of the transparency in detail.

Group B

Attempt Any Three Questions:

[3 x 10 = 30]

1. What is database and database management system? Explain the various database management system (DBMS) Component Modules in detail. [2 + 8]
2. What is join operation? List the various types of join operations? Explain the various methods used for implementing the join operation in query processing. [1 + 1 + 8]
3. What is query optimization? Write Heuristic Algebraic Optimization Algorithm. How the query optimizer transforms following query by using various heuristics into an equivalent optimized query that is more efficient to execute? Explain in detail. [1 + 3 + 6]

```
SELECT  E.LNAME
FROM    EMPLOYEE  E,      WORK_ON W,      PROJECT P
WHERE   P.PNMUBER = W.PNO
        AND  E.ESSN = W.SSN
        AND  E.BDATE > '1957-12-31'
        AND  P.PNAME = 'AQUARIUS' ;
```

4. What is distributed database? List advantages of distributed database. Explain data fragmentation, replication, and allocation techniques for distributed database design in detail. [1 + 3 + 6]

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Group A

Attempt Any Six Questions:

[6 x 5 = 30]

1. What is inheritance in an ODB? How can type inheritance and table inheritance be achieved in an ODB? Explain with suitable examples.
2. What is query processing? Explain the various steps involved in query processing in detail.
3. What is a nested subquery? How can nested subqueries be optimized? Explain in detail.
4. What is a distributed database? Explain the general architecture of a pure distributed database with a suitable diagram.
5. What is a NoSQL database? Explain why a NoSQL database system cannot guarantee all three properties of the CAP theorem simultaneously.
6. What is Big Data? Explain the major characteristics of Big Data using the 5Vs.
7. What is a deductive database? Explain how it differs from a conventional database.

Group B

Attempt Any Three Questions:

[3 x 10 = 30]

1. What is the conceptual design of a database? Compare the conceptual designs of ODB and RDB. Explain the steps involved in mapping an EER schema to an ODB schema in detail. [1 + 3 + 6]
2. Write the sort-merge algorithm for external sorting. Explain operator-level, intraquery, and interquery parallelism in query processing for parallel databases in detail. [4 + 6]
3. What is Incremental View Maintenance? Explain the Incremental View Maintenance technique for the relational operations of selection, projection, intersection, and join in detail. [2 + 8]
4. What are concurrency control and recovery? Explain various techniques for dealing with concurrency control and recovery in distributed database systems. [2 + 8]

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Group A

Attempt any three questions:

[63x 5 =10]

1. What is object Database? Compare the conceptual designs of object database (ODB) and relational database (RDB). Explain the steps involved in mapping an enhanced entity relationship (EER) schema to an ODB schema in detail. [1+3+6]
2. What is Incremental View Maintenance? Explain the Incremental View Maintenance technique for the relational operations of selection, projection, intersection, and join in detail. [2+8]
3. What is concurrency control and recovery? Explain various techniques for dealing with concurrency control and recovery in distributed database systems. [2 + 8]
4. What is NoSQL database? Why NoSQL database system cannot guarantee all three properties of the CAP theorem simultaneously. Explain CRUD operation for document based NOSQL database system in the context of MongoDB. [1+ 4+ 5]

Group B

Attempt any six questions:

[6 x 5 =30]

8. What is object and literal in object-oriented database? Compare object identifier (OID) with primary key. [2 + 3]
9. What is Set difference operation? Write Algorithms for processing query having Set difference operation.
10. What is a nested subquery? How can nested subqueries be optimized? Explain in detail.
11. What is federated database? Explain the Federated Database Schema Architecture with a suitable diagram.
12. What is wide column NOSQL systems? Explain CRUD operations in Hbase wide column NOSQL system. [1+4]
13. What is MapReduce? Explain MapReduce Programming Model with suitable example. [1+4]
14. What is a spatial database? Explain how it differs from a conventional database. [1+4]

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Group A

Attempt any three questions:

[63x 5=10]

1. What is join operation? Explain the various methods used for implementing the join operation in query processing. [1+9]
2. What is query optimization? Compare heuristic based and cost-based query optimization? How the query optimizer transforms following query by using various heuristics into an equivalent optimized query? Explain in detail. [1 + 3 + 6]

```
SELECT  E.LNAME
FROM    EMPLOYEE  E,      WORK_ON W,      PROJECT P
WHERE   P.PNMUBER = W.PNO
        AND  E.ESSN = W.SSN
        AND  E.BDATE > '1957-12-31'
        AND  P.PNAME = 'AQUARIUS' ;
```

3. What is distributed database? List advantages of distributed database. Explain data fragmentation, replication, and allocation techniques for distributed database design in detail. [1+2+7]
4. What is Hadoop? Explain the core components of the Hadoop ecosystem, including Hadoop MapReduce programming paradigm, Hadoop Distributed File System (HDFS), and YARN Resource Manager. Illustrate the working of MapReduce with a suitable example. [1+6+3]

Group B

Attempt any six questions:

[6 x 5 =30]

1. What is inheritance in an ODB? How can type inheritance and table inheritance be achieved in an ODB? Explain with suitable examples. [1+4]
2. What is query processing? Explain the various steps involved in query processing in detail. [1+4]
3. What is Union operation? Write Algorithms for processing query having Intersection operation. [1+4]
4. What is a distributed database? Explain the general architecture of a distributed database with a suitable diagram. [1+4]
5. What is document based NOSQL database system? Explain CRUD Operation in MongoDB. [1+4]
6. What is Big Data? Explain the major characteristics of Big Data using the 5Vs.[1+4]
7. What is temporal database? Explain design and Implementation Issues for temporal Databases in detail. [1+4]